The Relation Layer

Invariant cohomology of token-configuration 2-complexes: capacity defects, relation deficits, and predicate dependence

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Lead-in

Transformation without loss closed with five open items. This paper settles two of them and, in settling the second, changes what the third one is.

The first is arithmetic. The capacity ladder p_0, p_1, p_2, … records how many independent transformation channels a symmetric observer gains each time a distinguished site is allowed to hold one more token. That paper proved each rung is the first Betti number of a small quotient graph, proved a formula for the top rung, and observed that the whole ladder is a single function P_X(k) of the remaining occupancy k = K − r − 1. It asked for a recurrence or closed form and left it open. There is a closed form: one polynomial assembled from the cycle index of Stab(m) together with three integer statistics per group element. It never builds an arena, and at trivial site stabiliser it collapses to a quadratic times a binomial power.

The second is not arithmetic. The same paper recorded, as measured but not proved, that the invariant harmonic space vanishes — ε_harm = 0 across three arenas and both standard 2-cell predicates — and concluded that the entire invariant non-exact sector is coexact. It asked for a case with ε_harm ≠ 0, or a proof that none exists, naming this as the single result that would reopen a non-coexact channel, and nominating high-girth substrates with small automorphism groups.

ε_harm ≠ 0 occurs. The minimal witness is the bull — a triangle carrying a pendant on two of its three vertices, five vertices, |Aut| = 2. The nominated candidate class is wrong in a specific and useful way: girth is neither necessary nor a useful predictor in any family tested, and the surviving sector is not winding, because trees carry it. What forces it in a large and explicitly identifiable class of cases is a dimension obstruction that is vacuous on every arena previously tested, which is why it was never written down.

And then the question changes. The follow-up conjecture — that a substrate with at most two vertex orbits cannot carry the sector — survives 112 of 112 adversarial cases under one 2-cell predicate and

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fails under the other, on the same graph, the same group, the same invariant 1-skeleton and the same ε. Precisely:

$$H^1_G \text{ is not determined by } (X, K, G, C^1_G, d_0). \text{ It depends on } (C^2_G, d_1).$$

No function of the invariant 1-skeleton data alone can classify the harmonic sector across admissible relation predicates. That is the result the title refers to, and it is why the relation layer is named rather than the phenomenon.

A concrete instance of each, both reproduced in Section 16. On the 7-cycle with four tokens the published ladder runs 16 → 27 → 31 → 31; the closed form returns its increments 11, 4, 0 as three coefficients of one polynomial, checkable by hand. And the bull has two invariant non-exact channels and exactly one invariant 2-cell: one relation cannot close two channels, so one survives.

0. Scope and conventions

This is a mathematics paper. It belongs to a larger programme (NEXUS) whose ambition is explanatory rather than descriptive. That ambition appears in Section 18 and supports nothing. The rule followed throughout is the one used previously: prove downward, interpret upward, never prove upward by interpretation.

Three conventions are load-bearing.

(a) Every quantity involving d_1 carries its predicate. We write ε_harm(X, K, G, $\mathcal{K}$) throughout and suppress arguments only where all four are fixed in context. Section 12 shows this is not pedantry: a quoted ε_harm without its predicate can be either of two different numbers on the same graph.

(b) Every certificate of vanishing is exact. Integer matrices, modular elimination over two primes, rational cross-checks. Floating-point ranks appear only as diagnostics. An earlier stage of this programme once reported full rank on a matrix whose entries were of size 10^{-16} because a relative tolerance was used; a certificate of the shape given in Section 11 cannot fail that way.

(c) The generating function P_X(t) is a formal object whose coefficients equal capacity defects only in the non-degenerate range. Section 5.4 fixes the convention; in particular P_X(n) = −1 is a coefficient of a polynomial and is never a Betti number.

0.1 Status of every claim

The paper carries theorems, finite exact certificates, measurements and interpretation. They are separated below and the separation is maintained in the text.

Result	Where	Status
Splitting law for the capacity layer	Thm 3.1	Theorem
Burnside closed form for P_X(t)	Thm 4.1	Theorem
Trivial-site-stabiliser factorisation	Cor 5.1	Corollary

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Non-degenerate layers are cone-connected	Prop 5.5	Theorem
$\varepsilon_{harm} \neq 0$ occurs; the bull is minimal	Thm 8.1	Theorem + exhaustive certificate
Relation-deficit bound $\varepsilon_{harm} \geq \varepsilon - \dim C^2_G$	Thm 10.1	Theorem
Predicate monotonicity under inclusion	Thm 10.3	Theorem
Modular rank certificate	Prop 11.1	Proposition
Non-determination (Cor 12.1), 8-vertex witness	§12	Theorem + exact computation
112/112 all-4-cycle vanishing	§12	Finite exact verification
Vertex-orbit count as predictor	§14	Empirical, sampled population
Spectator-level profiles	§13	Measurement only
Pocket-capacity law $\min(K-1, r)$	§15	Measured in companion programme
NEXUS reading	§18	Interpretation

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1. Definitions

The definitions below are given in full because the central result of Part II is a statement about which of them the answer depends on. A reader should be able to reconstruct every computation from this section alone.

1.1 Arena

X = (V, E) is a finite connected simple graph, K a token count with $0 \le K \le |V|$. The arena $A_K(X)$ has as vertices the K-subsets $S \subseteq V$ and as edges the unordered pairs $\{S, S - u + w\}$ with $uw \in E$, $u \in S$, $w \notin S$.

Lemma 1.1 (edge parametrisation). Arena edges are in Aut(X)-equivariant bijection with pairs (e, S_0) where $e = uw \in E$ and $S_0 \subseteq V \setminus e$ with $|S_0| = K - 1$; the two orientations correspond to the two choices of endpoint carrying the token at the source. We call S_0 the spectator set.

This is Lemma 1.1 of the earlier work, restated because every Burnside computation in Part I runs through it. Note that it is the base graph, not the arena, that carries the indexing data.

1.2 Invariant cochains and reversible orbits

Fix $G \le \mathrm{Aut}(X)$. An oriented arena edge is an ordered pair (A, B) with {A, B} an arena edge. C^1 is the space of real functions on oriented arena edges satisfying $\omega(B, A) = -\omega(A, B)$; C^0 is the real functions on arena vertices; d_0 is $(d_0 f)(A, B) = f(B) - f(A)$. C^0_G and C^1_G denote the G-invariant subspaces.

An arena edge orbit is reversible if some $g \in G$ carries (A, B) to (B, A). Invariance then forces $\omega(A,B) = \omega(B,A) = -\omega(A,B)$, so ω vanishes on the whole orbit.

Lemma 1.2 (orientation count). Let G act on the arena edges. Each unoriented edge orbit lifts to exactly one oriented edge orbit if it is reversible, and to exactly two otherwise. Consequently

$$\dim C^1_G = \#\{\text{non-reversible unoriented orbits}\} = \#\{\text{oriented orbits}\} - \#\{\text{unoriented orbits}\}.$$

Proof. G acts on the two-element set of orientations of a fixed unoriented edge orbit through the quotient of the setwise stabiliser by the pointwise stabiliser, an action on two points. That action is transitive exactly when some element reverses the edge, giving one oriented orbit; otherwise it is trivial on each orientation class, giving two. Summing over unoriented orbits gives the displayed identity, and the first equality is Theorem 2.1 of the earlier paper. ■

The second equality is what makes reversibility Burnside-countable. Reversibility is not a property of a single pair (e, S_0) — it depends on the whole group — so it cannot be counted by a fixed-point sum directly; the difference of two fixed-point sums evades this.

Finally $\varepsilon(X, K, G) := \dim C^1_G - \dim (\mathrm{im}\, d_0)^G$. Averaging over the finite group gives $(\mathrm{im}\, d_0)^G = d_0(C^0_G)$, so $\varepsilon = \dim C^1_G - \mathrm{rank}(d_0 \mid C^0_G)$.

1.3 The two 2-cell predicates

A 2-cell is a closed 4-cycle of arena edges, recorded as a cyclic sequence (A, B, C, D) of four distinct arena vertices with AB, BC, CD, DA arena edges, taken up to cyclic rotation; the reversed sequence is the opposite orientation of the same 2-cell. A predicate $\mathcal{K}$ is a G-stable set of such 2-cells.

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$\mathcal{K}$_disj (disjoint-support commuting squares). For each K-subset S and each ordered pair of base edges $e_1 = u_1w_1$ and $e_2 = u_2w_2$ with $e_1 \cap e_2 = \emptyset$, $u_1, u_2 \in S$ and $w_1, w_2 \notin S$, the four configurations S, $S - u_1 + w_1$, $S - u_1 - u_2 + w_1 + w_2$, $S - u_2 + w_2$ in that cyclic order. Each such square is recorded once.

$\mathcal{K}_4$ (all arena 4-cycles). Every 4-cycle of the arena 1-skeleton, each recorded once up to rotation and reversal.

Lemma 1.3. $\mathcal{K}$_disj $\subseteq \mathcal{K}_4$.

Proof. The four configurations of a disjoint-support square are pairwise distinct, because e_1 and e_2 are disjoint, and consecutive ones differ by a single legal move. ■

The inclusion is strict in general; on Q_3 at K = 3 the counts are 168 and 306. Lemma 1.3 is used in Theorem 10.3.

1.4 Invariant 2-cochains

C^2 is the space of real functions on oriented 2-cells of $\mathcal{K}$ that change sign under reversal of the cyclic order. For $\omega \in C^1$ define

$$(d_1\,\omega)(A, B, C, D) = \omega(A,B) + \omega(B,C) + \omega(C,D) + \omega(D,A)\,,$$

the signed circulation of ω around the cell. C^2_G is the G-invariant subspace. Exactly as for edges, a 2-cell orbit is reversible if some $g \in G$ carries the cell to itself with the opposite cyclic orientation, in which case every invariant 2-cochain vanishes on it; hence

$$\dim C^2_G = \#\{\text{non-reversible } G\text{-orbits of 2-cells in } \mathcal{K}\}\,.$$

This is the same rule that defines dim C^1_G, applied one dimension up, and it is stated explicitly because Theorem 10.1 depends on the number it produces.

Lemma 1.4. $d_1\, d_0 = 0$ for any predicate $\mathcal{K}$.

Proof. $(d_1\, d_0\, f)$ evaluated on (A,B,C,D) is [f(B)−f(A)] + [f(C)−f(B)] + [f(D)−f(C)] + [f(A)−f(D)] = 0. The cells of $\mathcal{K}$ are closed edge-cycles, so the sum telescopes regardless of which cycles are chosen. ■

The identity therefore follows from the definition rather than from a numerical check; the check reported in Section 16 is a test of the implementation, not of the mathematics.

1.5 Harmonic cohomology

Give C^0_G, C^1_G and C^2_G the Euclidean inner products in which the orbit bases described above are orthonormal, and let d_1^* denote the adjoint of d_1 with respect to these. Define

$$\mathcal{H}^1_G := \ker(d_1 \mid C^1_G) \cap (\operatorname{im} d_0)^{\perp}\,, \quad \varepsilon_harm := \dim \mathcal{H}^1_G\,.$$

By Lemma 1.4 and finite-dimensional Hodge theory,

$$C^1_G = \operatorname{im} d_0 \oplus \mathcal{H}^1_G \oplus \operatorname{im} d_1^*\,, \quad \text{and} \quad \varepsilon_harm = \dim H^1(C\bullet_G) = \dim \ker d_1 / \operatorname{im} d_0\,.$$

The two descriptions have equal dimension and are used interchangeably only for that reason; $\mathcal{H}^1$_G and the quotient $H^1(C\bullet_G)$ are not the same object. Terminology: exact means in im d_0, coexact means in im d_1^*, and harmonic means closed and coclosed, hence orthogonal to both. The earlier phrase "the

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invariant non-exact sector is coexact" is the statement $\mathcal{H}^1_G = 0$; a harmonic class is not merely non-coexact, and that looser reading is avoided here.

Write $B^1_G = \mathrm{im}(d_0 \mid C^0_G)$ and let $\bar{\partial}_1 : C^1_G / B^1_G \to C^2_G$ be the induced map, well defined by Lemma 1.4. Then

$$\varepsilon_harm = \varepsilon - rank(\bar{d}_1) = dim\ C^1_G - rank\ d_0 - rank\ d_1, \quad and \quad \varepsilon_harm = 0 \Leftrightarrow \bar{d}_1\ is\ injective.$$

All of Part II is a statement about $\mathrm{rank}(\bar{\partial}_1)$.

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Part I. The capacity ladder

2. The capacity layer and the cone

Fix a distinguished site m ∈ V, write G = Stab_{Aut(X)}(m), and let $X^{(r)}$ be the arena in which m may hold up to r tokens and every other site at most one, with K tokens in total. Then $X^{(r)} \subset X^{(r+1)}$, the restriction σ_r on invariant residuals is surjective, and the capacity defect is D_r = ker σ_r with p_r = dim D_r.

The newly admitted layer consists of the configurations with occupancy exactly r + 1 at m:

$$\mathcal{D}_r = \{\, m^{r+1} + U : U \subseteq V \setminus \{m\},\ |U| = k \,\}, \quad k = K - r - 1 .$$

Write Y = X − m for the deleted substrate and n = |V(Y)| = |V| − 1. The restriction of G to Y is faithful: an element of Stab(m) fixing every vertex of Y fixes m as well, hence is the identity. (Connectedness of X is not needed for this; it is needed later, in Proposition 5.5.)

Proposition 2.1 (layer structure). As a G-set, $\mathcal{D}$_r is the vertex set of the k-token arena A_k(Y). The arena edges of $X^{(r+1)}$ meeting $\mathcal{D}$_r are of exactly two kinds:

- internal: a move u ⟶ v with u, v ≠ m. These preserve the occupancy r + 1 and are precisely the edges of A_k(Y);
- cross: a move m ⟶ v with v ∈ N(m) \ U, which lowers the occupancy to r and therefore lands in $X^{(r)}$.

Proof. A configuration of the layer is determined by U since the multiplicity at m is fixed at r + 1, and G fixes m, so the identification is equivariant. A legal move either leaves the occupancy at m unchanged, which forces both endpoints to lie in V \ {m}, or changes it by one. A move into m would give occupancy r + 2, which exceeds the capacity of $X^{(r+1)}$ and is therefore not an edge of that arena. ■

A remark on the asymmetry, because it is easy to misread. The underlying token dynamics is not directed. Cross edges are one-sided only after restricting attention to the top newly admitted stratum: within $X^{(r+1)}$ a cross edge is an ordinary undirected arena edge with one endpoint in $\mathcal{D}$_r and one in $X^{(r)}$. What Proposition 2.1 says is that no arena edge leaves $\mathcal{D}$_r upward, because upward is outside the arena.

Since the endpoints of a cross edge have different occupancies at m and G fixes m, no cross orbit can ever be reversible. This is the general reason the published top-layer formula carries no reversibility correction, where it had been established by inspection of a single case.

The cone quotient $\boldsymbol{y}$_r is the multigraph whose vertices are the G-orbits of $\mathcal{D}$_r together with one additional vertex ∗ representing the collapsed $X^{(r)}$, whose edges are the G-orbits of internal moves (joining two layer orbits) and of cross moves (joining a layer orbit to ∗), and from which the reversible orbits have been deleted. Distinct v ∈ N(m) \ U give distinct arena edges with the same pair of endpoints in the quotient, so $\boldsymbol{y}$_r is genuinely a multigraph and parallel edges must be counted with multiplicity. Writing

E_nr(k) for the number of non-reversible G-orbits of edges of A_k(Y); V_orb(k) for the number of G-orbits of k-subsets of V(Y); and x_k for the number of G-orbits of pairs (U, v) with |U| = k and v ∈ N(m) \ U, we have E_Q = E_nr(k) + x_k and V_Q = V_orb(k) + 1.

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3. The splitting theorem

Theorem 3.1 (splitting). Let X be connected, m ∈ V, G = Stab(m), and let the layer be non-degenerate, that is $X^{(r)} \neq \emptyset$. Then

$$p_r = P_X(k) = E_nr(k) + x_k - V_orb(k)\,, \quad k = K - r - 1\,.$$

Proof. The cone theorem of the earlier paper gives p_r = $\beta_1(\mathcal{Y}_r)$ = E_Q − V_Q + c_Q, and Proposition 5.5 below gives c_Q = 1 whenever $X^{(r)} \neq \emptyset$. Substituting E_Q and V_Q from Section 2, the two additive constants cancel. ■

Two features deserve emphasis. First, no connectivity hypothesis is imposed on Y. This matters: A_k(Y) is genuinely disconnected whenever m is a cut vertex — up to six components in the tested cases — and yet

$$A_k(Y) \text{ disconnected } \not\Rightarrow \mathcal{Y}_r \text{ disconnected}\,,$$

because the cone point reconnects the components. Routing the internal part through the published residual formula ε = E − V + c introduces a component correction that must then be removed again; see Section 17, correction (ii). Second, at k = 0 the layer is the single configuration m^K, so E_nr = 0, V_orb = 1 and x_0 = #(N(m)/G), recovering the published top-layer formula D_{K−1} = #(N(m)/Stab(m)) − 1 as a special case.

4. The Burnside closed form

Each of the three counts in Theorem 3.1 is a Burnside sum, and each collapses — not by algebraic accident, but because of which cycles of g the defining constraint touches. For g ∈ G write

$$Z_g(t) = \prod_{c \in cyc_g(V(Y))} (1 + t^{|c|})\,,$$

the subset cycle index, and define three integers: a(g), the number of edges of Y fixed pointwise by g; b(g), the number of edges of Y whose two endpoints g interchanges; f(g), the number of neighbours of m fixed by g.

Configuration orbits. A k-subset is g-invariant precisely when it is a union of cycles of g, so by Burnside ∑_k V_orb(k) t^k = $|G|^{-1}$ ∑_g Z_g(t).

Cross orbits. A pair (U, v) is fixed by g precisely when g(v) = v and g(U) = U with v ∉ U. But g(v) = v makes {v} a singleton cycle of g, and the admissible U are exactly the g-invariant subsets of V(Y) \ {v}, whose generating function is Z_g(t) with that singleton factor removed. Hence ∑_k x_k t^k = $|G|^{-1}$ ∑_g f(g) Z_g(t)/(1 + t).

Edge orbits. By Lemma 1.2, E_nr is the difference of an oriented and an unoriented orbit count, and by Lemma 1.1 an arena edge is a pair (base edge, spectator set of size k − 1 disjoint from it), which accounts for the overall factor t. An oriented such pair is fixed by g iff g fixes both endpoints of the base edge; they are then two singleton cycles, and the spectator generating function is $Z_g/(1 + t)^2$. An unoriented pair is fixed iff g fixes the base edge setwise, which happens either pointwise — again $Z_g/(1 + t)^2$ — or by interchanging the endpoints, in which case those two vertices form a 2-cycle of g and the spectator generating function is $Z_g/(1 + t^2)$. Subtracting, the pointwise contribution survives once and the swapped contribution enters with a minus sign.

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Theorem 4.1 (closed form). Let X be connected, m ∈ V, G = Stab(m), Y = X − m. Then

$$P_X(t) = |G|^{-1} \sum_{g \in G} Z_g(t) \left[a(g)\cdot t/(1+t)^2 - b(g)\cdot t/(1+t^2) + f(g)/(1+t) - 1 \right],$$

and p_r = [t^{K−r−1}] P_X(t) for every non-degenerate layer.

Every division is exact. If a(g) > 0 then g has two fixed points, so $(1 + t)^2$ divides Z_g; if b(g) > 0 then g has a 2-cycle, so $(1 + t^2)$ divides Z_g; if f(g) > 0 then g has a fixed point, so $(1 + t)$ divides Z_g. Terms with a zero coefficient are absent and impose no divisibility requirement.

No arena is constructed at any point. The cost is one pass over G, recording a cycle index and three integers per element. The pair a(g), b(g) with its minus sign is precisely the "cycle structure of g on V plus two edge counts per group element" of the published Theorem 2.3, recovered here in generating-function form; the f(g) term is new and is the only place the distinguished site enters.

5. Consequences

5.1 Trivial site stabiliser.

Corollary 5.1. If Stab(m) is trivial then, with n = |V| − 1, e = |E(Y)| and d = deg m,

$$P_X(t) = (1+t)^{n-2} \left[(d - 1) + (e + d - 2)\, t - t^2 \right].$$

Proof. $Z(t) = (1 + t)^n$, a = |E(Y)|, b = 0, f = d; substitute and factor. ■

Since Y is X with m and its d incident edges removed, e + d = |E(X)|, so the quadratic rewrites with no reference to Y:

$$P_X(t) = (1+t)^{|V|-3} \left[(\deg m - 1) + (|E(X)| - 2)\, t - t^2 \right].$$

Coefficientwise, with B_k = C(n − 2, k), the ladder is P_X(k) = (d−1)B_k + (e+d−2)B_{k−1} − B_{k−2}, a three-term relation inherited from Pascal's. That is the recurrence the earlier open list asked for.

The corollary has a negative consequence worth isolating. At trivial site stabiliser the closed-form ladder depends on X through exactly three integers: |V|, |E(X)| and deg m. It does not depend on degree sequence, triangle count, girth or finer connectivity data. Among the 19 connected graphs on six vertices with seven edges, eight non-isomorphic pairs (X, m) with degree sequences ranging from [1,1,2,3,3,4] to [2,2,2,2,3,3] and triangle counts 0, 1 and 2 carry the identical ladder [1, 8, 17, 13, 2, −1]. This is the sharp form of the earlier boundary note that ε "is not a base-graph invariant": at trivial site symmetry the ladder is a base-graph invariant, and a very weak one.

5.2 Cycles through a reflection.

For X = C_N the site stabiliser is {1, σ}, with σ the reflection through m. With n = N − 1 even, σ has n/2 two-cycles on Y = P_n, exactly one swapped edge, no fixed vertices and no fixed neighbours of m, so

$$P(t) = \tfrac{1}{2} \left[(1+t)^{n-2}(1 + (n-1)t - t^2) - (1+t^2)^{n/2-1}(1 + t + t^2) \right].$$

For C_7 this is $4t + 11t^2 + 14t^3 + 6t^4 - t^6$, whose t^2, t^1 and t^0 coefficients are 11, 4 and 0 — the published ladder 16 → 27 → 31 → 31 at K = 4.

5.3 K-independence.

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K does not appear in P_X(t). The earlier paper observed that p_r depends only on k and wrote P_X on that basis; Theorem 4.1 makes it structural, since the layer as a G-set depends only on k and the cross structure only on N(m).

5.4 The degenerate corner, and a convention.

At k = n the layer is the single saturated configuration: one vertex orbit, no internal edge because there is no vacancy, and no cross edge because N(m) ⊆ U. The formula returns 0 + 0 − 1 = −1 for every substrate, site and group. By Proposition 5.5 that is exactly the corner where $X^{(r)}$ is empty and no capacity transition exists.

Convention. p_r = P_X(k) holds on the non-degenerate range only. The value P_X(n) = −1 is a coefficient of the formal generating function and is not a Betti number, a dimension, or a capacity defect. We keep it because it is a useful detector: over 433 (substrate, site, k) points, the only negative coefficients occur at k = n and every one of them equals −1, so a negative coefficient identifies the degenerate corner exactly.

5.5 Particle–hole symmetry, and where it breaks.

Complementation S ↦ V \ S makes ε particle–hole symmetric, as proved in the earlier work. The distinguished site breaks that symmetry, and at trivial site stabiliser the closed form says by exactly how much:

$$P_X(n-k) - P_X(k) = \deg(m)\,[\,C(n-1,\,k-1) - C(n-1,\,k)\,]\,.$$

The asymmetry is carried entirely by deg m, the one place the substrate touches the capacity site.

5.6 No non-degenerate disconnected layer.

Proposition 5.5. Let X be connected and let the layer be non-empty with $X^{(r)} \neq \emptyset$. Then $\mathcal{Y}$_r is connected, so c_Q = 1.

Proof. Because X is connected, every component of Y contains at least one neighbour of m. Let U be a layer configuration. If some component Y_i of Y satisfies |U ∩ Y_i| < |Y_i| then that component holds a vacancy; since Y_i is connected, the vacancy can be moved through Y_i by internal moves until some neighbour of m lying in Y_i is unoccupied, and the resulting configuration admits a cross move to ∗. If instead |U ∩ Y_i| = |Y_i| for every i then k = n, the degenerate corner, where $X^{(r)}$ is empty. Reversible orbits are deleted from $\mathcal{Y}$_r, but a reversible orbit is a loop in the quotient and deleting loops does not change connectivity. ■

This settles item 4 of the earlier open list for the present admissibility rules, replacing a 492-transition search by an argument, and it covers cut vertices, where A_k(Y) itself is disconnected.

6. Reach

The value of a closed form is the cases the arena cannot reach. P_X(t) returns the whole ladder for every K at once at the cost of one pass over Stab(m). On the 5 × 6 grid — thirty vertices, with trivial stabiliser at the chosen corner site — the polynomial is produced in under a millisecond and gives P(14) = 945,414,540, for a layer whose configuration count is C(29,14) = 77,558,760. On C_{32} it gives P(15) =

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1,168,225,605 against a layer of C(31,15) = 300,540,195 configurations. Direct computation of either is out of the question.

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Part II. The relation layer

7. ε is 1-skeleton data

Part I lives entirely on the invariant 1-skeleton. Part II does not, and that distinction is the content of this half.

The two predicates of Section 1.3 share C^0_G, C^1_G, d_0 and therefore ε exactly — none of those objects mentions a 2-cell. They differ in C^2_G and d_1, and therefore may differ in $\mathcal{H}^1_G$. Section 12 exhibits a case where they do. The statement to keep in view is:

ε is 1-skeleton data; ε_harm is relation-layer data.

8. Existence

Two things must hold before any nonzero result can carry weight. The published zeros must come back, and the implementation must realise the complex of Section 1: by Lemma 1.4, $d_1d_0 = 0$ is automatic, so a nonzero measurement of d_1d_0 would signal a coding error in the orientation bookkeeping. Both check out: the zeros on C_8, Q_3 and Petersen reproduce under both predicates, max $|d_1d_0| = 0$ exactly across sixteen complexes, and Euler characteristic consistency holds 16/16.

Theorem 8.1 (existence and minimality). ε_harm ≠ 0 occurs. The bull — a triangle with a pendant attached to two of its three vertices — has |V| = 5, |Aut| = 2, ε = 2 and ε_harm = 1 under both predicates at K = 2 and K = 3. No connected graph on four vertices gives ε_harm ≠ 0 at any K under either predicate; of the twenty-one connected graphs on five vertices exactly three do, all with |Aut| = 2, and the bull is the one with fewest edges.

Minimality is by exhaustion over isomorphism classes; existence for the bull is a finite exact computation. This answers item 2 of the earlier open list in the affirmative and reopens the non-coexact channel.

The revision forced on Section 2 of the earlier paper is narrow and exact: the invariant non-exact sector is coexact in a restricted high-symmetry regime, not in general. C_8, Q_3 and Petersen are vertex-transitive and lie inside that regime; they are not generic evidence.

9. What does not explain it

Three candidate explanations fail. The null results are the useful part, because each removes an address.

Girth.

The nominated candidate class was high girth with small automorphism group. In the tested families high girth is neither necessary for nonvanishing nor a useful sufficient predictor of it. Petersen at girth 5, C_8 with a pendant tail at girth 8 with |Aut| = 2 and ε = 50, and C_9 at girth 9 all give ε_harm = 0, while the minimal witness has girth 3. No high-girth substrate tested — including the subdivisions constructed adversarially for Section 12, which are high-girth by design — carries the sector.

Winding.

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Trees carry it. The caterpillar on seven vertices has $\beta_1(X) = 0$ and gives $\varepsilon_harm = 1$ at $K = 2$ and 2 at $K = 3$; an asymmetric spider on eight vertices gives 0 at $K = 2$ and 1 at $K = 3$. A substrate with no cycle supports no substrate winding, so the surviving sector is not winding under another name. It is multi-token configuration structure — tokens exchanging around essential vertices — which is why one-body reasoning does not see it.

Marginality.

The sector is not a sliver at the edge of measurement. Two 4-cycles joined by a bridge and carrying three pendants, with trivial automorphism group, give $\varepsilon_harm = 7$ at $K = 2$ and 12 at $K = 3$. In the exhaustive six-vertex population 98 of 224 (substrate, K) pairs are nonzero, with maximum 8.

10. The relation-deficit theorem

Theorem 10.1 (invariant relation deficit). For any X, K, G and predicate $\mathcal{K}$,

$$\varepsilon_harm = \varepsilon - rank(\bar{d}_1) \geq \varepsilon - dim\ C^2_G\ , \quad hence \quad \varepsilon > dim\ C^2_G \Rightarrow \varepsilon_harm > 0\ .$$

Proof. By Lemma 1.4 the map $\bar{d}_1 : C^1_G/B^1_G \to C^2_G$ is well defined and $\varepsilon_harm = \dim(C^1_G/B^1_G) - \text{rank}(\bar{d}_1) = \varepsilon - \text{rank}(\bar{d}_1)$. The rank of a linear map is at most the dimension of its codomain, so $\text{rank}(\bar{d}_1) \leq \dim C^2_G$. ■

In words: if the symmetry-surviving non-exact channels outnumber the symmetry-surviving relations available to constrain them, at least one channel must remain both closed and non-exact. The bound is elementary, and it is vacuous on C_8, Q_3 and Petersen, where the invariant 2-cells comfortably outnumber ε — which is why it was never written down in a programme whose test arenas were all of that type.

The formulation through $\bar{d}_1$ rather than $d_1 \mid C^1_G$ matters: ε has already removed $\text{im}\ d_0$, and the two ranks agree only because $d_1 d_0 = 0$. An equivalent bookkeeping is $\varepsilon_harm = \dim C^1_G - \text{rank}\ d_0 - \text{rank}\ d_1$.

The theorem resolves the minimal examples mechanically rather than descriptively:

Graph	ε	dim C^2_G	lower bound	measured ε_harm
paw	0	0	0	0
bull	2	1	1	1
wheel W_6	3	3	0	0

The paw has no invariant non-exact direction at all. The bull has two and exactly one invariant 2-cell: one relation cannot eliminate two independent directions, so at least one survives, and the computation finds exactly one. W_6 has three channels and three relations — the count permits complete closure, and closure occurs. The bull is therefore not merely the smallest graph on which a harmonic class was observed; it exhibits the mechanism in one line.

Theorem 10.2 (coverage). In the exhaustive population of the 112 connected graphs on six vertices at K = 2 and K = 3 — 222 complexes under $\mathcal{K}$_disj — there is no violation of Theorem 10.1. The bound is active in 85 cases, all 85 of which have ε_harm > 0, with no false positives, and it is exactly

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tight in 61. Of the 98 nonzero cases observed in this population it certifies 85, that is 87% of the observed nonzero cases in this exhaustive six-vertex population, by counting alone and with no rank computation.

The remaining 13 are qualitatively different: there $\varepsilon \le \dim C^2_G$, so counting cannot decide whether ∂_1 is injective. They turn on dependence among the relation coordinates, which a dimension count cannot see. Theorem 10.1 is a sufficient condition for nonvanishing, not a characterisation.

Theorem 10.3 (predicate monotonicity). If $\mathcal{K} \subseteq \mathcal{K}'$ are two predicates on the same arena and group, then $\varepsilon_harm(X, K, G, \mathcal{K}') \le \varepsilon_harm(X, K, G, \mathcal{K})$.

Proof. $\mathcal{K}'$ imposes every relation that $\mathcal{K}$ imposes and possibly more, so $\ker(d_1^{\mathcal{K}'} \mid C^1_G) \subseteq \ker(d_1^{\mathcal{K}} \mid C^1_G)$, while $\operatorname{im} d_0$ is unchanged. Intersecting both with $(\operatorname{im} d_0)^{\perp}$ gives $\mathcal{H}^1_G(\mathcal{K}') \subseteq \mathcal{H}^1_G(\mathcal{K})$. ■

With Lemma 1.3 this gives $\varepsilon_harm(\mathcal{K}_4) \le \varepsilon_harm(\mathcal{K}_disj)$ always, so the two predicates are ordered rather than merely different, and the direction of the counterexample in Section 12 is forced rather than accidental. Verified over the same 224 complexes with no violation and strict inequality in 64 of them. This is a partial answer to the question of which predicate pairs can disagree: predicates ordered by inclusion disagree only in one direction.

11. Certifying zero

Theorem 10.1 only ever proves $\varepsilon_harm > 0$. Vanishing is the opposite estimate — ∂_1 must be injective — and the bound is structurally silent there. An exact certificate is nevertheless available.

Work in the orbit basis of C^1_G, of dimension $t = \dim C^1_G$. Let M_0 be the matrix of d_0 restricted to invariant 0-cochains and M_1 the matrix of d_1 on that basis; both have integer entries, since every structure constant is 0 or ±1 by construction.

Proposition 11.1 (modular rank certificate). Let p be a prime. If $\operatorname{rank}_p M_0 + \operatorname{rank}_p M_1 = t$ then $\varepsilon_harm = 0$.

Proof. By Lemma 1.4, $\operatorname{im} M_0 \subseteq \ker M_1$, so over $\mathbb{Q}$ we have $\operatorname{rank}_{\mathbb{Q}} M_0 + \operatorname{rank}_{\mathbb{Q}} M_1 \le t$, and $\varepsilon_harm = t - \operatorname{rank}_{\mathbb{Q}} M_0 - \operatorname{rank}_{\mathbb{Q}} M_1 \ge 0$. For any prime, $\operatorname{rank}_p M \le \operatorname{rank}_{\mathbb{Q}} M$, since a nonvanishing minor mod p lifts to a nonvanishing minor over $\mathbb{Q}$. If the modular ranks already sum to t then

$$t \le \operatorname{rank}_{\mathbb{Q}} M_0 + \operatorname{rank}_{\mathbb{Q}} M_1 \le t\,,$$

so both rational ranks are determined and $\varepsilon_harm = 0$. ■

The certificate is one-sided: a modular sum below t proves nothing, because rank can drop mod p. Every vanishing result reported in Sections 12 and 16 is obtained this way, with two primes (2147483647 and 2147483629) as an audit layer and a sample cross-checked by exact rational elimination. The failure mode recorded in Section 8 of the earlier paper — a relative tolerance reporting full rank on a matrix of entries of size 10^{-16} — cannot arise in a computation of this shape.

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12. The eight-vertex counterexample

With existence controlled by Theorem 10.1, the remaining question is which substrates force vanishing. An empirical pattern presented itself and looked strong: among the substrates tested at that stage — exhaustive on six vertices, sampled on seven, plus named larger graphs — every substrate with at most two vertex orbits gave ε_harm = 0, in 86 of 86 cases, including W_6 and the pentagonal prism, which carry ε = 1, 11 and 24. Vanishing was therefore not an artefact of an empty residual.

The pattern is false. An adversarial search was run — cones over vertex-transitive graphs, which carry large ε; subdivisions, which are simultaneously two-orbit and high-girth, exactly the class the original candidate hypothesis pointed at; complete multipartite graphs; windmills and books; and random two-orbit graphs on seven, eight and nine vertices. It breaks at eight.

The counterexample has |V| = 8, |Aut| = 8, vertex orbits {0,2,5,6} and {1,3,4,7}, degree sequence 4,4,4,4,5,5,5,5, and edge set

01 02 03 05 07 12 14 15 23 24 26 36 37 45 46 56 57 67.

Its exact rational ranks are these.

K	predicate	t	rank$_{\mathbb{Q}}$ M_0	rank$_{\mathbb{Q}}$ M_1	sum	ε	ε_harm
2	$\mathcal{K}$_disj	14	6	7	13 / 14	8	1
2	$\mathcal{K}_4$	14	6	8	14 / 14	8	0
3	$\mathcal{K}$_disj	32	9	22	31 / 32	23	1
3	$\mathcal{K}_4$	32	9	23	32 / 32	23	0

Same graph, same group, same invariant 1-skeleton, same ε. The harmonic dimension is 1 or 0 according to which 4-cycles are declared to be relations.

Across the adversarial pool of substrates with at most two vertex orbits, under the certificate of Proposition 11.1:

predicate	cases	ε_harm = 0	counterexamples
$\mathcal{K}$_disj	112	104	8
$\mathcal{K}_4$	112	112	0

So the vanishing statement is not a theorem about a graph together with its symmetry. For $\mathcal{K}$_disj it is false, with a minimal counterexample at eight vertices. For $\mathcal{K}_4$ it survives every adversarial case tested, each one certified rather than observed — but that is a finite verified statement about a tested population, not a universal theorem, and the distinction is drawn deliberately, because the 86/86 that preceded it looked equally convincing.

Corollary 12.1 (non-determination). The invariant 1-skeleton data do not determine the harmonic sector across admissible relation predicates:

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$(X, K, G, C^1_G, d_0) \not\Rightarrow H^1_G$ across admissible relation predicates.

Equivalently: there is no function $F(X, K, G, C^1_G, d_0)$ whose value determines ε_harm across the predicates $\mathcal{K}_disj$ and $\mathcal{K}_4$.

Proof. The two complexes built on the graph above share all five arguments and have different ε_harm. ■

The displayed form is the one to quote. It states the obstruction at the level of the data rather than of a hypothetical classifier, and it does not presuppose which relation system is the correct one: whatever is computed from the invariant 1-skeleton alone, two admissible predicates over it can disagree. By Theorem 10.3 that disagreement is at least directed.

The classifier must see the relation layer.

13. The spectator-degree grading

The arena cochain groups carry a filtration by spectator dependence. Via Lemma 1.1 an arena edge is a pair (base edge e, spectator set S), and F_p is spanned, for each base edge, by the indicators $\mathbb{1}[T \subseteq S]$ with $|T| \le p$. We call this the spectator-degree filtration; its graded pieces are governed by the Johnson-scheme decomposition of functions on k-subsets. We avoid the term "Johnson filtration", which has an established and different meaning in the theory of mapping class groups.

Level 0 is not a new object. It is exactly the reader space $\pi^*C^1(X)$ of Theorem 3.1 of the earlier paper: a matrix-reader cochain depends only on the moved base edge and its orientation, which is degree zero in S. That gives an independent anchor, and the published Section 9 values return exactly, including the one nontrivial invariant part.

substrate	K	dim F_0	published $\pi^*C^1(X)$	invariant part	published
C_8	3	8	8	0	0
C_8	4	8	8	0	0
Q_3	3	12	12	0	0
Petersen	3	15	15	0	0
P_8	3	7	7	3	3

With the grading anchored, the harmonic sector can be resolved by level. These are reported as measurement only, for the reason given in Section 17, correction (iv): bull [0,1]; C_5 with two pendants [0,1]; caterpillar [0,1] at K = 2 and [0,1,1] at K = 3; asymmetric spider [0,0] and [0,0,1]; two bridged 4-cycles [2,5] and [2,5,5]. No claim in this paper rests on them.

14. Empirical predictors

Two scalar predictors were tested against the exhaustive six-vertex population. Both are recorded at the strength the data support, which is less than either initially appeared to carry.

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Automorphism-group order is not a control parameter. The hit rate falls with |Aut| — 16/16, 57/74, 20/56, 2/14, 1/18, 2/20, then zero at every order of 16 and above — but the observed bin frequencies are not monotone, the value at order 12 exceeding that at order 8.

Vertex-orbit count is monotone in the same data where |Aut| is not: 0/8, 0/36, 17/54, 31/66, 34/42, 16/16 across one to six orbits, and 0/2, 0/4, 17/66, 135/192, 208/250, 268/278, 246/246 on a seven-vertex sample. It is a useful empirical predictor in the sampled population, and the vertex-transitive corner explains in one line why the three originally tested arenas gave zero.

It is not the mechanism. Section 12 exhibits a graph with two vertex orbits whose harmonic dimension changes when only the 2-cells change; a fixed orbit count therefore does not determine ε_harm. The mechanism is the rank of the declared invariant relation map ∂_1 on the residual.

The three-orbit population, which looked like a phase boundary, resolves accordingly: of its 17 nonzero cases, 13 are forced by Theorem 10.1 and 4 lie below the counting threshold. It is not one boundary but a mixture of dimensionally forced and rank-sensitive sectors.

15. A companion measurement: pocket capacity

The following is recorded for completeness because it bears on the same distinction and is referenced in the open list. It is a measurement from the companion coarse-arena programme and was not re-run for this paper; it is reported as measured there, with its falsification test still outstanding.

In a coarse arena that is not a valid model of the underlying configuration space, a cyclic gadget region can behave like a finite storage pocket and generate spurious classes indexed by how many spectators are held in it. The measured rank of the affected region obeys

$$r_base(K) = min(K - 1, r)\,, \quad r_tail(K) = min(K - 1, \ell)\,,$$

with r the number of vertices in the gadget region and ℓ the length of a genuine tail. The measured values are:

Edge gadget	region r	base rank, K = 2,3,4,5	tail rank, K = 2,3,4,5
one p on triangle edge, one tail vertex, no q	2	1, 2, 2, 2	1, 1, 1, 1
original p–q region plus tail	3	1, 2, 3, 3	1, 1, 1, 1
original region with tail length 2	4	1, 2, 3, 4	1, 2, 2, 2
q-to-base path length 2	4	1, 2, 3, 4	1, 1, 1, 1

The two inequivalent r = 4 constructions agree through K = 5, which indicates that the observed quantity depends on the number of locally available storage vertices and not on the detailed topology of the gadget. The outstanding falsification test is r = 4 at K = 6, where the law predicts r_base(6) = min(5,4) = 4; a rank of 5 would refute it. That computation requires on the order of 7,700 invariant orbits and has not been run.

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The relevance here is the same one Section 12 makes: two models can produce the same counts for different reasons. A genuine tail is an obstruction to rearrangement; a coarse cyclic pocket is an artefact of insufficient subdivision. Equal counts are not equal mechanisms.

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16. Verification

All computation is self-contained: itertools, collections, numpy and fractions only, with no imports from earlier artifacts. Automorphism groups, arenas, both 2-cell complexes, the invariant Hodge decomposition, the spectator-degree filtration and exact/modular rank are built from scratch. Two independent implementations of ε — orbit counting and explicit rank — are carried throughout and asserted equal at every call.

16.1 Reproduction of published values.

```
eps(C8, K=3,4,5)      = [2, 3, 2]         [paper: [2, 3, 2]]
eps(C_N, K=3) N=5..9   = [0, 0, 1, 2, 3]    [paper: [0, 0, 1, 2, 3]]
eps(K_N, all K)       = [0, 0, 0, 0, 0]    [paper: all 0]
eps(Q3,3)=1  eps(Petersen,3)=1  eps(P8,3)=24   [paper: 1, 1, 24]


C7  m=0  K=4  |Stab(m)|=2
  r=1->2: dim R 16->27  coker=0  D_r=11 | Q_r: V=10 E=20 c=1 beta1=11   OK
  r=2->3: dim R 27->31  coker=0  D_r=4  | Q_r: V=4  E=7  c=1 beta1=4    OK
  r=3->4: dim R 31->31  coker=0  D_r=0  | Q_r: V=2  E=1  c=1 beta1=0    OK
Q3  m=0  K=4  |Stab(m)|=6
  r=1->2: dim R 25->36  coker=0  D_r=11 | Q_r: V=7  E=17 c=1 beta1=11   OK
  r=2->3: dim R 36->39  coker=0  D_r=3  | Q_r: V=4  E=6  c=1 beta1=3    OK
  r=3->4: dim R 39->39  coker=0  D_r=0  | Q_r: V=2  E=1  c=1 beta1=0    OK
prism  m=0  K=4  |Stab(m)|=2
  r=1->2: dim R  8->20  coker=0  D_r=12 | Q_r: V=7  E=18 c=1 beta1=12   OK
  r=2->3: dim R 20->25  coker=0  D_r=5  | Q_r: V=4  E=8  c=1 beta1=5    OK
  r=3->4: dim R 25->26  coker=0  D_r=1  | Q_r: V=2  E=2  c=1 beta1=1    OK
P6 (interior site)  m=2  K=4  |Stab(m)|=1
  r=1->2: dim R  6->17  coker=0  D_r=11 | Q_r: V=11 E=21 c=1 beta1=11   OK
  r=2->3: dim R 17->23  coker=0  D_r=6  | Q_r: V=6  E=11 c=1 beta1=6    OK
  r=3->4: dim R 23->24  coker=0  D_r=1  | Q_r: V=2  E=2  c=1 beta1=1    OK
ALL PUBLISHED ROWS REPRODUCED: True
```

16.2 Part I.

```
splitting law  P_X(k) = E_nr(k) + x_k - V_orb(k)     438/438 transitions
Burnside pieces (E_nr, V_orb, x_k) vs direct counting 1299/1299
closed-form ladder vs measured capacity defect       1026/1026
non-degenerate layers with c_Q > 1                   0
closed form on cut-vertex sites                      85/85
binomial formula (trivial site stabiliser)           552/552
factorisation (1+t)^(|V|-3) * quadratic              44/44
```

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K-independence: D_r depends on (X,m,k) only 171/171

sign law: 433 points, 68 negative, all at k=n, all = -1

particle-hole defect = deg(m)*[C(n-1,k-1)-C(n-1,k)] 264/264

C7 closed form P(2),P(1),P(0) = 11, 4, 0 [paper: 11, 4, 0]

C9 ladder [0, 6, 26, 58, 67, 44, 12, 0, -1]

C11 ladder [0, 8, 47, 148, 268, 312, 226, 100, 20, 0, -1]

C13 ladder [0, 10, 74, 300, 735, 1216, 1376, 1084, 570, 190, 30, 0, -1]

cross-substrate collision, (n,|E(X-m)|,deg m) = (5,5,2): 8 non-isomorphic
(X,m) with degseq [1,1,2,3,3,4] .. [2,2,2,2,3,3], triangles 0/1/2,
all with ladder P(k) = [1, 8, 17, 13, 2, -1]

5x6 grid (30 vertices): closed form 0.3 ms; P(14) = 945,414,540
that layer alone has C(29,14) = 77,558,760 configurations

C_32: P(15) = 1,168,225,605 vs C(31,15) = 300,540,195

16.3 Part II.

published zeros reproduced, both predicates:
C8 K=3 cells=80/80 eps=2 eps_harm=0
C8 K=4 cells=120/120 eps=3 eps_harm=0
Q3 K=3 cells=168/306 eps=1 eps_harm=0
Petersen K=3 cells=450/450 eps=1 eps_harm=0

implementation check: max |d1 . d0| over 16 complexes = 0.000e+00

Euler consistency b0-b1+b2 == dimC0-dimC1+dimC2 : 16/16

exhaustive minimality:
n=4: 6 iso classes, 0 witnesses
n=5: 21 iso classes, 3 witnesses, all |Aut|=2

substrate	b1(X)	girth	|Aut|	K	eps	eps_harm
bull	1	3	2	2	2	1
bull	1	3	2	3	2	1
paw	1	3	2	2	0	0
Petersen	6	5	120	3	1	0
C9	1	9	18	3	3	0
C8+tail	1	8	2	3	50	0
caterpillar TREE	0	inf	2	2	6	1

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caterpillar TREE 0 inf 2 3 14 2

asym spider TREE 0 inf 2 3 24 1

two C4s + pendants 2 4 1 2 54 7

two C4s + pendants 2 4 1 3 268 12

Theorem 10.1, exhaustive 6-vertex, 222 complexes, K_disj:

violations 0

active (eps > dim C2_G) 85 of those eps_harm>0: 85

exactly tight when active 61/85

coverage of nonzero cases 85/98 (87%)

Theorem 10.3, same 224 complexes:

K_disj subset of K_4 : True

eps_harm(K_4) <= eps_harm(K_disj) : True, 0 violations

strict inequality : 64 cases

eps identical under both predicates : all checks equal

hit rate by vertex-orbit count (6-vertex exhaustive):

1 orbit 0/8 2 orbits 0/36 3 orbits 17/54

4 orbits 31/66 5 orbits 34/42 6 orbits 16/16

16.4 The predicate split, exact certification.

pool: 60 substrates with at most 2 vertex orbits

(cones over vertex-transitive graphs carry large eps; subdivisions are

2-orbit AND high-girth, the class the original conjecture pointed at)

K_disj : 104/112 certified eps_harm=0 counterexamples: 8

rand8_0 |V|=8 K=2 eps=8 eps_harm=1

rand8_0 |V|=8 K=3 eps=23 eps_harm=1

rand8_1 |V|=8 K=2 eps=11 eps_harm=1

rand8_1 |V|=8 K=3 eps=29 eps_harm=1

rand8_3 |V|=8 K=2 eps=11 eps_harm=1

rand8_3 |V|=8 K=3 eps=29 eps_harm=1

rand8_4 |V|=8 K=2 eps=11 eps_harm=1

rand8_4 |V|=8 K=3 eps=29 eps_harm=1

K_4 : 112/112 certified eps_harm=0 counterexamples: 0

counterexample, EXACT RATIONAL rank (floating point not used):

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|V|=8 |Aut|=8 orbits=[[0,2,5,6],[1,3,4,7]] degseq=[4,4,4,4,5,5,5,5]

K=2 K_disj t= 14 rank_Q(M0)= 6 rank_Q(M1)= 7 sum=13/14 eps= 8 eps_harm=1

K=2 K_4 t= 14 rank_Q(M0)= 6 rank_Q(M1)= 8 sum=14/14 eps= 8 eps_harm=0

K=3 K_disj t= 32 rank_Q(M0)= 9 rank_Q(M1)=22 sum=31/32 eps=23 eps_harm=1

K=3 K_4 t= 32 rank_Q(M0)= 9 rank_Q(M1)=23 sum=32/32 eps=23 eps_harm=0

17. Corrections made during the derivation

Each of these is a distinction the final statements depend on. Four are hypotheses of mine that the data killed before they reached a claim; they are recorded because the surviving results are the ones that survived them.

(i) The ordinary-arena group action.

The arena was first implemented with configurations encoded as (occupancy at m, set of other occupied sites), and that encoding was applied with the full automorphism group. It is valid only for Stab(m), which preserves the occupancy split; a general automorphism moves m. Every capacity row still matched, because those use Stab(m), but ε(C_N, K=3) came out [2, 4, 8, 12, 17] against the published [0, 0, 1, 2, 3]. The earlier published dataset functioned as an independent regression test and caught the bug. The ordinary arena now uses plain subsets.

(ii) A spurious component term in the splitting law.

Theorem 3.1 was first written as P_X(k) = ε(Y,k) + x_k − c_k, routing the internal part through the published residual ε = E − V + c and then correcting for the components of A_k(Y). That double-counts. The sweep failed on exactly the cut-vertex cases, each off by c_k − 1. The component term cancels identically.

(iii) Killed: monotonicity in |Aut|.

An earlier reading described the ε_harm hit rate as monotone in automorphism-group order. The observed bin frequencies are not monotone; see Section 14. Retired.

(iv) Killed: harmonic support confined to level p ≥ 1.

From the bull, the pendant cycles and the trees — all of which profile [0, 1, …] — it appeared that a nontrivial automorphism group forbids level-0 harmonic content. Tested exhaustively on six vertices, 72 of 82 symmetric cases carry level-0 content. The rule held only on the hand-picked family. This is why Section 13 reports profiles as measurements and builds nothing on them.

(v) Killed: at most two vertex orbits forces vanishing.

86 of 86 supporting cases, then an adversarial search broke it at eight vertices. Retained only in the weakened, predicate-qualified form of Section 12. The failure is what forced the separation of the 1-skeleton from the relation layer, which is the main result of Part II.

(vi) The original candidate class for ε_harm ≠ 0.

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The earlier open list nominated high-girth substrates with small automorphism groups. The small-group half pointed usefully. The high-girth half is contradicted by every high-girth substrate tested, including the subdivisions built for the adversarial pool.

18. Boundary of the result

These are not disclaimers. Each records a distinction established during the derivation that prevents two mathematically different objects from being identified because they agreed on a slice of the parameter space.

P_X(k) is not a physical quantity. It is a sequence of Betti numbers in the non-degenerate range and a formal polynomial coefficient outside it. No dimensional map is constructed, and the boundary section of the earlier paper stands unchanged: "price", mass and information loss remain unavailable as readings.

ε_harm is not a force or a curvature channel. It is dim $\mathcal{H}^1$_G under a declared predicate. Its nonvanishing means there exist invariant directions orthogonal to both the exact and the coexact subspaces of that complex. Naming that a physical channel requires an independently defined map, which is not constructed here.

ε belongs to the 1-skeleton; ε_harm does not. Reporting ε_harm without its predicate is not a stylistic lapse: Corollary 12.1 exhibits one graph, one group and one invariant 1-skeleton on which the value is 1 or 0 depending on the declaration.

Theorem 10.1 is sufficient, not necessary. It forces nonvanishing in a large and explicitly identifiable class; 13 of the 98 observed nonzero six-vertex cases lie below its threshold.

The all-4-cycle vanishing result is finite. 112 of 112 tested cases are certified exactly. That is not the universal theorem, and the 86/86 that preceded its $\mathcal{K}$_disj counterpart is the reason for insisting on the distinction.

The NEXUS reading is interpretation.

With that boundary in place the shape is worth stating once, in its technical form first. The harmonic sector is

$$\mathcal{H}^1_G = \ker(d_1 : C^1_G \to C^2_G) \cap (\mathrm{im}\, d_0)^{\perp}.$$

Read through the programme's vocabulary: the invariant 1-skeleton records which transformations are available, the declared 2-cells record which combinations of transformations count as relations, and the symmetry quotient records which distinctions remain independently representable. Theorem 10.1 then says that when the available invariant channels outnumber the admitted invariant relations, complete closure is impossible. Corollary 12.1 says that the relation layer cannot be determined from the tested 1-skeleton data alone — which is weaker, and more defensible, than saying it cannot be derived from the transformation layer at all: there might well be a canonical construction producing a particular C^2 from further structure on X or from an externally specified theory. What is established is that no such construction is implicit in (X, K, G, C^1_G, d_0). The linear algebra is proved; the ontology is not thereby proved, and nothing in Sections 1–16 depends on it.

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19. Open

The remaining problem is not a hunt for a graph invariant. It is a question about invariant chain complexes:

Characterise when $\bar{d}_1 : C^1_G / \mathrm{im}\, d_0 \to C^2_G$ is injective.

- Prove or refute: for $\mathcal{K}_4$, at most two vertex orbits implies ε_harm = 0. The $\mathcal{K}$_disj counterexample shows such a theorem must use the structure of the all-4-cycle relation system; it cannot follow from vertex-orbit count alone.
- Theorem 10.3 orders predicates related by inclusion. Characterise disagreement for predicates that are not nested, and determine whether $\mathcal{K}_4$ is maximal among predicates on a given arena.
- The 13 nonzero six-vertex cases that Theorem 10.1 does not certify. There ε ≤ dim C^2_G and the answer turns on dependence among the relation coordinates.
- Determine whether ε_harm resolved by spectator degree connects to the exchange-level hierarchy of the companion papers. The grading is anchored at level 0; the profiles of Section 13 are measurements awaiting a theorem.
- Run r = 4 at K = 6 against the pocket-capacity law of Section 15. The prediction is r_base(6) = 4; a rank of 5 refutes it.
- From the earlier list, still open: the one-sided exactness statements, and the reproduction or retraction of the excluded multigraph and filled-square-complex tables. Theorem 4.1 makes the first cheaper, since candidates can be screened by P_X(t) and ε without building an arena.

Reproducibility

Environment. Python 3.11.15, numpy 2.4.4, Linux x86-64. Standard library: itertools, collections, fractions. No other dependencies.

Certificates. Modular ranks are computed by exact Gaussian elimination over F_p with p = 2147483647 and p = 2147483629, agreement over both primes required; a sample is cross-checked by exact rational elimination over $\mathbb{Q}$ using Python fractions. Floating-point ranks, where reported as diagnostics, use an absolute singular-value tolerance of 10^{-9}; a relative tolerance is never used.

Canonicalisation. Automorphism groups are computed by backtracking over degree-compatible vertex assignments with adjacency consistency checked against all previously assigned vertices; orbit representatives are the least element of each orbit under the vertex order. Isomorphism-class deduplication in the exhaustive sweeps is by minimum over all vertex permutations of the sorted edge set.

Graph encodings. Every named substrate in this paper is specified by an explicit edge list in the companion notebooks; the eight-vertex counterexample is given in full in Section 12. The random two-orbit substrates are regenerated deterministically from a fixed seed, recorded in the notebook.

Predicates. $\mathcal{K}$_disj and $\mathcal{K}_4$ are enumerated exactly as defined in Section 1.3; both enumerations are in the companion notebooks and each 2-cell is recorded once up to rotation and reversal.

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Every number in Sections 2–16 marked as measured is produced by the companion notebooks, which run top to bottom from a single self-contained library cell. Licence CC BY-NC 4.0. Repository github.com/QuHarmonics.

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